NATIONAL UNIVERSITY



Fourth Year Syllabus

Department of Library and Information Science (LIS)

Four-Year B.A. (Honours) Course

Effective from the Session: 2013–2014

NATIONAL UNIVERSITY

Syllabus for Four Year B.A. Honours Course
Subject: Library and Information Science (LIS)
Session: 2013-2014

FOURTH YEAR

Paper Code	Paper Title	Marks	Credits
243801	Organization of Knowledge (Classification Theory)	100	4
243803	Organization of Knowledge (Cataloguing Theory)	100	4
243805	Systems Analysis and Design	100	4
243807	New Technologies Current Trends in Information Systems	100	4
243809	Information Production Marketing and Public Relation	100	4
243811	Information Retrieval Techniques	100	4
243813	Advanced Bibliography	100	4
243814	Organization of Knowledge (Classification Practical)	75	3
243816	Organization of Knowledge (Cataloguing Practical)	75	3
243820	Computer Applications in Libraries (Practical)	50	2
243822	Viva Voce	100	4
	Total =	1000	40

OBJECTIVES

(1) To give the students an understanding of the basic principles of fundamental laws of Library and Information Science and to enable them to understand and appreciate the functions, purposes and services of Libraries and Information Organizations in this fast changing world of 'Information Age'.

(2) To train the students in the techniques of Information Management and equip them with the latest developments in Information Technology (IT) and its applications in libraries and information centres.

(3) To acquaint the students with the organizations and development of the Universe of Knowledge.

Detailed Syllabus

Paper Code	243801	Marks: 100	Credits: 4	Class Hours: 60 hrs.
Paper Title:	**Organization of Knowledge (Classification Theory)**			

Unit-1 Meaning and purpose of organization of library materials

Unit-2 Theory of classification; natural and artificial classification; knowledge classification; principles of book classification; rules of subject determination

Unit-3 Special features of library classification: study of different kind of library materials including specialized and non-book materials

Unit-4 Characteristics of classification schemes: enumerative and analytico-synthesis schemes

Unit-5 Major schemes of library classification:

a) Dewey Decimal Classification (DDC): history and origin, principles, collocation, seven tables, notations, number building in DDC.
b) Universal Decimal Classification (UDC): growth and development of the scheme, auxiliaries in UDC, use of its application in special libraries.
c) Library of Congress Classification (LC): growth and development, principles of the schemes, its application in libraries.

Unit-6 Problems of library re-classification.

Reading list:

Broadfield, A. The philosophy of classification.
Datta, D.N. Library Classification Manual.
Kellay, G.D. The classification of books and inquiry into its usefulness to the readers.
Mil, J. A. Modern outline of library classification.
Perrsault, J. M. Towards a theory for UDC.
Philips, W. H. A printer of books classification.
Ranganathan, S. R. Prolegomena to library classification.

Paper Code	243803	Marks: 100	Credits: 4	Class Hours: 60 hrs.
Paper Title:	Organization of Knowledge (Cataloguing Theory)			

Unit-1 Modern concept of cataloguing procedures

Unit-2 Major cataloguing codes: ALA 1949, AACR 1967, AACR2R: 1988

Unit-3 Subject determination: use of Sears List, direction for use, principles of construction, types of subject headings, construction of subject headings with subdivisions

Unit-4 Shelf list: definition, functions and importance; shelf list vs. public catalogue; accession register.

Unit-5 Union catalogue: definition, functions and importance; kinds and compilation; procedures of union catalogue; use and implementation of union catalogue in Bangladesh

Unit-6 Analytical entry: definition, kinds, functions and importance

Unit-7 Cataloguing entries: definition, kinds, functions and importance; Indexing of catalogue entries; rules for arranging index entries

Unit-8 Organization of technical section; cataloguing tools: needs and significance; behind the screen process

Unit-9 Computerized catalogues: hardware and software for integrated library automation systems; input and output devices; MARC programme; development of CD-ROM, OPAC, etc.

Reading list :

Akers, S. G. Simple library cataloguing,
Hunter E. and Bakewell, K.G. Cataloguing,
Hunter, E. Computerized cataloguing.
Margaret, M. Introduction to cataloguing and classification of books.
Needham, C. D. Organizing knowledge in libraries: an introduction to information retrieval. Redfren, B. Cataloguing.
Sengupta, B. Cataloguing: its theory and practice.
Wyner, B. S. Introduction to cataloguing and classification.

Paper Code	243805	**Marks: 100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title:	**Systems Analysis and Design**			

Unit-1 Introduction to information systems: system concepts, information system and its characteristics; elements and types; computer as a cybernetic system; system analysts and their role, responsibilities, essential qualification and professional issues

Unit-2 System analysis: system requirements, tools for planning and developing of information system, System Development Life Cycle (SDLC), understanding existing system, plan for system study, information gathering, definition of problem and specific objectives, search for alternatives, feasibility study, functional steps, evaluation and selection of a candidate system, writing feasibility report, operational goal, specific requirement and evaluation criteria of new candidate system

Unit-3 System design: process and methodologies, design background, patterns and modules of system design, input design, control design, output design, database design, data structure, data communication and network design, program and procedure design, hardware and software specifications

Unit-4 Implementation: Implementation plan, method of testing, evaluation, installation and selection of hardware and software, recruitment and training of personnel. system conversion, post-implementation follow-up, evaluation and maintenance

Unit-5 Cost-benefit analysis: concepts, cost and benefit categories, study of cost salaries, space, supplies and inventors, overhead implement costs, investment costs, study of tangible and intangible benefits, procedure for cost-benefit determination

Unit-6 Programming: conceptual issues, objectives, functional steps in programming, programming techniques, algorithm, flowcharting, branching and looping, debugging, introduction to some programming language, Basic, Pascal, Java, HTML

Unit-7 Evaluation of information system and services: importance and scope of evaluation in library and information systems, considering issues for evaluating an information system, performance measurement-objectives, different phases of performance measurement, Evaluation of systems security and data integrity

Reading list:

Award, E. M. Systems Analysis and Design.
Hawry S. I. System Analysis and Design.
Rajaraman, V Analysis and Desgn.
Whitten, J. L., Bently, L. D. and Dittmen, K. C. System Analysis and Design Concept.

Paper Code	243807	**Marks: 100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title:	**New Technology Current Trends in Information Systems**			

Unit-1: Concept of information communication technologies (ICTs), evolution and development of ICT, use and applications of ICTs in different fields of library and information institutions in Bangladesh, impact of ICT in library and information systems, current trends and existing situation, problems and prospects of using ICT in Bangladesh

Unit-2: Computer, concepts, types of computer based on processing/signaling, purpose and capacity of size, comparisons of mainframe, mini- and microcomputers, generation of computers, CPU Control unit, arithmetic logic unit, primary memory comparison of primary and secondary memory

Unit-3: Digital library, concepts, necessity, function, characteristics, major activities and skills or digital librarianship, digital library scenario in Bangladesh

Unit-4: Computer hardware, overview of computer hardware, basic components of a computer system, input and output components of a computer system, factors to be considered for the purchase of hardware of micro computers in library and information centre/institution, distinction between color monitor and monochrome monitor

Unit-5: Software, concepts, classification of software and introduction of some application software used for the library and information centre in Bangladesh

Unit-6: Operating system, concepts, types of operating system based on processing and user interface, functions of operating system, introduction to some popular operating systems, process, process management

Unit-7: Information superhighway: definition, components, user groups, importance, advantages and disadvantages of information superhighway; World Wide Web, Internet, web browser (Internet Explorer, Netscape Navigator, etc.); web pages andwebsites

Unit-8: Computer networks, concepts, types of computer networks and their configurations, layers and protocol of computer networks.

Reading list:

Minasi, M. The Complete PC Upgrade and Maintenance Guide.
Peter, N. Introduction to Computer.
Peter, N. Inside the PC.
Rahman, M. L. Hossain, M.A. Computer Fundamentals.
Redd, L. A. Introduction to Computer-Based Library Systems.

Paper Code	243809	**Marks: 100**	**Credits: 4**	**Class Hours: 60 hrs.**
Paper Title:	**Information Production Marketing and Public Relation**			

Unit-1: Information, identification of information need and information seeking behavior, needs, wants and demands, product, products and services, product classifications, individual product decisions, information as a product, concept of fee based information services, awareness and promotion of information products and services, designing information products and services for users, current trends in Bangladesh

Unit-2: Marketing concept, marketing defined, key purposes of marketing, organizational orientations towards marketing, factors affecting achievement or organization library objectives, barriers to marketing, marketing management developing marketing programs for libraries, marketing approach marketing and the quality revolution, market, what can be marketed, market planning and implementation status of the use of marketing concepts in libraries and information centres, need for strategic involvement, professional skills for marketing, marketing problems in libraries and information centres, benefits of marketing

Unit-3: Developing marketing plan: What is marketing plan benefits of a marketing plan, how to develop a marketing plan, marketing audit, strategic direction for information centre, implementation and control, marketing mix, four Ps and four Cs, creating the marketing mix

Unit-4: Market segmentation: Characteristics of segmentation, levels of segmentation, methods of segmentation requirements for effective segmentation. Marketing communications: Communication process, steps in developing effective communication

Unit-5: Direct and online information marketing: Direct marketing, benefits an growth of direct marketing, forms of direct marketing. Online marketing of information products in service. Conducting online marketing, challenges of online marketing, challenges of online marketing: new roles of libraries and information on professionals, marketing libraries and information centres in the digital world, the digital marketing mix

Unit-6: Marketing research and information systems: Assessing information needs, developing and distributing information. Marketing research: defining the problems and research objectives, preparing the research brief, developing the research plan, description of research designs

Unit-7: Reprography: conceptual issues, types of reproduction, different methods and process of reprography, reprography, reprographic technologies, planning for a reprographic unit, reprographic services in Bangladesh

Unit-8: Advertising, promotion and public relations: Setting advertising objectives, developing advertising strategy, other advertising considerations, Public relations planning AIDA-a communication model, professional cooperation and promotion, the media, book talk, direct marketing, press releases and press conferences

Unit-9: E-commerce and its application in library and information services

Reading list:

Elliott de S. E. Marketing concepts for libraries and information services.
Jain, A. K. and Others. Marketing information products and services: a primer for librarian and information professional.
Kotler P. and Armstrong, G. Principles of marketing.
Blaise, G. The marketing of library and information services.
Irving, A. Marketing the information profession to the information society.

Paper Code	243811	Marks: 100	Credits: 4	Class Hours: 60 hrs.
Paper Title:	**Information Retrieval Techniques**			

Unit- 1 Information retrieval: definitions, types and tools for information retrieval; different methods of information retrieval (IR), processes of IR, current technologies and traditional methods towards IR; evolution of IR systems

Unit-2 Stages of information retrieval: planning, analysis, implementation and evaluation

Unit-3 The nature of retrieval: indexing, storage, retrieval; indexing and searching Languages; search Logic; basic retrieval facilities, search strategies; search management

Unit-4 User profile: definitions, characteristics, elements, creating user profile in practice

Unit-5 Computer-based information retrieval system

Unit-6 Evaluation of information storage and retrieval systems, recall and precision, effectiveness, main causes of failure, cost-effectiveness, cost-benefit evaluation

Unit-7 Dissemination of information: forms of dissemination, dissemination of primary products, dissemination of secondary information, SDI

Reading list:

Guha, B. Documentation and Information retrieval.
Sengupta, B. and Chatterjee, M. Documentation and Information Retrieval.
Sharp, J. Information retrieval: Notes for Students.
Bunch, A. The Basics of Information Work.
Lancastester, F. W. Information Retrieval System.
Rowley, J. E. and Turner, C.M.D. The Dissemination of Information.

Paper Code	243813	Marks: 100	Credits: 4	Class Hours: 60 hrs.
Paper Title:	Advanced Bibliography			

Unit-1: Bibliography: definition, bibliography as a science or an art, importance of bibliography, bibliography vs. library catalogue, bibliography vs. footnotes, reference vs. bibliography

Unit- 2: Different types of bibliographies: analytical, historical, systemic, national, trade, subject, universal, bibliography of Bibliographies; incunabula Bibliographies, Author Bibliographies, etc.; National Bibliography of Bangladesh

Unit-3: Evaluation of bibliography: authority, scope, items and entries of information, special features, format, drawbacks, conclusion

Unit-4: Arrangement of different kinds of bibliography with proper examples

Unit-5: Bibliographic Organizations: FID, IFLA, ICSU, ISO, UNISIST, UNESCO, INSDOC, BANSDOC

Unit-6: Bibliographic Control: Different methods and policies of bibliographic control. Its usefulness in Bangladesh

Unit-7: Bibliographic Description: Minimum standard entry, Short standard entry, Full entry with examples

Unit-8: Bibliomatric: origin, growth, applications and limitations

Unit-9: Bibliotheraphy: importance in various treatment, Purposes. Bibliographic network. ISBN.

Unit-10: Literature search: Techniques, benefits, steps, citation analysis, Literature searching tools and their uses.

Reading list:

Chakraborti, M. L.; Bibliography in Theory and Practice.
Kumar, G. and Krishan K. Bibliography.
Kumar, K. Reference Service.
Ranganathan, S.R.; Reference Service.
Shores, L. Basic Reference Sources.
Winchell, C. M. Guide to Reference Books.

Paper Code	243814	**Marks: 75**	**Credits: 3**	**Class Hours: 60 hrs.**
Paper Title:	Organization of Knowledge (Classification Practical)			

An Introduction to DDC (19th edition): First Summary, Second Summary, Third Summary and others use and application of schedule (index and table), Number analysis, Number building with the aid of tables and schedules.

Paper Code	243816	**Marks: 75**	**Credits: 3**	**Class Hours: 60 hrs.**
Paper Title:	**Organization of Knowledge (Cataloguing Practical)**			

Analysis and practical application of cataloguing rules; AACR-2, the Sears List of Subject Headings (13th edition), and Library of Congress subject headings, Call Numbers, etc.

Indention and spacing on skeleton card: Title and statement of responsibility area; edition area; material (or type publication) of responsibility area; publication, distribution, etc.

Main entries: Single, double, triple, more than triple authors; Editor, compiler, translator, etc., Corporate bodies; Scared books; Title on main entry (Serial, Dictionary, and Encyclopedia)

Added Entries: Subject, Title, Joint Authors, Series, Editor, Compiler, Translator, etc. and Filing the entries according to dictionary and classified methods.

Reading list:

Batty, CD. An Introduction to 19th edition of DDC.
Melvil D. & Albany, N. Y. Dewey Decimal Classification and relative index.
John P. Comarami and others Manual of the use of the Decimal classification.
Saiful I. K.M. Number building in Dewey Decimal Classification, 19th and 16th editions, a practical manual.

Paper Code	243818	**Marks: 50**	**Credits: 2**	**Class Hours: 60 hrs.**
Paper Title:	**Computer Applications in Libraries (Practical)**			

MS-Word, Excel, PowerPoint and other office applictions
Library Software including CDS/ISIS, Koha, DSpace, Greenstone and other repository software.

Practical Examinations

Question papers for the practical examinations must be prepared by two examiners.
The practical questions must be moderated by the Examinations Committee like other papers/Papers.
Practical Paper teachers of respective department/college/institution together with the external examiners will arrange the practical examinations.
Marks of the practical examination must be submitted to the controller of examinations/Head of the Examinations Committee immediately at the end of the practical examinations.

Paper Code:	243820	**Marks: 100**	**Credits: 4**	
Paper Title:	**Viva-voce**			